Twenty-third day of class - 11/13

Fun fact: Let $E : y^2 = x^3+69x+22$. If p is a prime and $E(\mathbb{F}_p)$ contains a subgroup isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, then $E(\mathbb{F}_p)$ contains a subgroup isomorphic to $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$.

Last time: Elliptic curve cryptography. Elliptic curves over finite fields. An example of the quadratic sieve.

Today: The baby-step-giant-step algorithm for solving the elliptic curve discrete logarithm problem, attacks on the classical discrete logarithm problem, and the ellipitic curve factorization method.

The baby-step-giant step algorithm

Suppose that $E/\mathbb{F}_p$ is an elliptic curve, $P \in E(\mathbb{F}_p)$ is a point with order N. We're given P and aP and we want to find a.

Let $x = \lceil \sqrt{N} \rceil$. There are c_1 and c_2 with $0 \leq c_1 \leq x-1$ and $0 \leq c_2 \leq x-1$ so that

$$a = c_1 x + c_2.$$

(This is a base x representation of a.)

We make a list of values of $c_2 P$ for $0 \leq c_2 \leq x-1$.

We make a list of values of $aP - c_1 xP$ for $0 \leq c_1 \leq x-1$.

There must be a match between these two lists. If

$$aP - c_1 xP = c_2 P,$$

then $aP = c_1 xP + c_2 P$ and so $a = c_1 x + c_2$ works.

Do you want to see an example with Magma?

What about the classical discrete logarithm problem?

Given g and h in $(\mathbb{Z}/p\mathbb{Z})^\times$, the goal is to find x so that $g^x \equiv h \pmod{p}$.

There is an analogue of the quadratic sieve (called the "index calculus") for the classical discrete logarithm problem, and the existence of this algorithm makes it so key sizes for the classical Diffie-Hellman cryptosystem must be large. The idea is to find a multiplicative relation of the form

$$g^s h \equiv (-1)^{f_0} 2^{f_1} 3^{f_2} \cdots p_r^{f_r} \pmod{p}.$$

There is also a DLP analogue of the Number Field Sieve.

What about the elliptic curve discrete logarithm problem?

There is no analogue of the quadratic sieve known for the EC discrete logarithm problem. The reason is there is no natural notion of a "prime element" in $E(\mathbb{F}_p)$. In 1999, Silverman wrote a paper called "The Xedni Calculus" which was an attempt to attack the ECDLP. It involved taking a curve $E/\mathbb{F}_p$ and lifting the curve to a curve $E/\mathbb{Q}$ and doing computations there (in particular computing the rank of $E(\mathbb{Q})$) and attacking in this way. It is difficult to analyze in a theoretical way, and it appears to be very impractical.

The elliptic curve factorization algorithm (1987)

Short summary: If n is not prime, $\mathbb{Z}/n\mathbb{Z}$ is not a field. If we pretend it is, and do a bunch of elliptic curve arithmetic, we might get to a point where there is a number $a \in \mathbb{Z}/n\mathbb{Z}$, $a \neq 0$ whose multiplicative inverse we cannot compute. In that case $1 < \gcd(a, n) < n$ and we have factored n.

Method: Let n be a composite number.

1. Check that $\gcd(n, 6) = 1$ and n is not a perfect power.
2. Choose random integers b, x_1 and y_1 mod n.
3. Let $c = y_1^2 - x_1^3 - bx_1$ mod n and

$$C : y^2 = x^3 + bx + c.$$

Let $P = (x_1, y_1)$.

4. Check that $\gcd(4b^3 + 27c^2, n) = 1$.
5. Pick a number B_1 and let k be the least common multiple of 1, 2, 3, …, B_1.
6. Compute $kP = \left(\frac{a_k}{d_k^2}, \frac{b_k}{d_k^3}\right)$ and hope for an error.

(In practice, we do this by writing $k = \prod_{j=1}^{r} p_j^{e_j}$. Then we let $P_0 = P$, $P_1 = p_1^{e_1} P_0$, and $P_n = p_n^{e_n} P_{n-1}$.)

The goal is to have a prime p dividing n with the property that $P \in E(\mathbb{F}_p)$ has order a multiple of k, but for there to be another number q so that $P \in E(\mathbb{F}_q)$ does not have order a multiple of k. If that's the case, then the denominator of the x-coordinate of kP will be a multiple of p, but not of q.

7. If we don't get an error, go back to step 2 and try another curve.

Why does this work?

We are searching for a situation where the order of P mod p has only smallish prime factors (that is, the order of P divides k), but the order of P mod q does not.

The order of P divides the order of $|E(\mathbb{F}_p)|$ which is between $p + 1 - 2\sqrt{p}$ and $p + 1 + 2\sqrt{p}$. The hope is to randomly select a curve $E/\mathbb{F}_p$ so that $|E(\mathbb{F}_p)|$ divides k.

Example: (Chosen carefully)

Let $n = 10^{50} + 27$.

Let $\sigma = 3650126150$. Let $u = \frac{\sigma^2 - 5}{4\sigma}$. Let $a = (1/u - 1)^3(3u + 1)/4 - 2$. Let $x = u^3$, $b = x^3 + ax^2 + x$. Let

$$E : y^2 = x^3 + abx^2 + b^2 x,$$

and take $P = (bu^3, b^2)$. (This construction is due to Suyama and ensures that E has a point of order 12 on it.)

Let $B_1 = 17500$. We define $P_0 = P$, and for each prime p_i, $1 \le p \le B_1$, let $e_i = \lfloor \frac{\log(K)}{\log(p_i)} \rfloor$ and define $P_i = p_i^{e_i} P_{i-1}$.

When we compute this (in Sage), we find get an error that we cannot compute the multiplicative inverse of 9434472662078741753716090810630777612914159795355 mod n. This is because the gcd of this number with n is

$$p = 258706694329115968764 1.$$

The reason is that $E/\mathbb{F}_p$ has a group of order $2^2 \cdot 3 \cdot 17 \cdot 43 \cdot 67 \cdot 109 \cdot 193 \cdot 983 \cdot 12497 \cdot 17033$, which is a divisor of k. Thus, $kP = (0 : 1 : 0) \in E(\mathbb{F}_p)$.

However, if $q = 386538122870466317455356449 47$, then $|E(\mathbb{F}_q)|$ is a multiple of the large prime 354663393563136731 and the order of $P \in E(\mathbb{F}_q)$ is also a multiple of that prime. This makes it so that $kP \ne (0 : 1 : 0) \in E(\mathbb{F}_q)$, so the denominator of the x-coordinate of kP is a multiple of p but not of q.

You can reproduce this in Magma with the following commands:

```
n := 10^50 + 27;

ECM(n,17500 : Sigma := 3650126150);
```

In usual implementations, there is a second stage which requires more memory and allows one to compare the differences of x-coordinates of points computed to see if they have a common factor with n. If so, we also get a factorization.

Recommendations about the size of B_1 in comparison with the size of the prime factor sought.

Digits in factor	Optimal B_1	Average number of tries needed
20	1100	74
25	5000	214
30	25000	430
35	10^6	904
40	$3 \cdot 10^6$	2350
45	$11 \cdot 10^6$	4480
50	$43 \cdot 10^6$	7553
55	$11 \cdot 10^7$	17769
60	$26 \cdot 10^7$	42017
65	$85 \cdot 10^7$	69408

Record: In September of 2013, Ryan Propper found an 83 digit prime factor of $7^{337} + 1$ using ECM. (As a consequence, the full factorization of $7^{337} + 1$ is known.) He had $B_1 = 7.6 \cdot 10^9$. It took about 14 hours to check one curve, and Ryan did about 5000 curves before finding the factor (even though the estimate is that one million curves were necessary).

How do I use it?

The website `http://www.alpertron.com/ar/ECM.HTM` has an online calculator that factors numbers using both MPQS and ECM (as appropriate). It can be kind of fun to watch it while it works.

Examples to try:

1. $10^{80} + 73$. The website tries ECM and finds one prime factor. After dividing out by that prime the number is still composite, and it uses the quadratic sieve to finish. (Total time is about 15 seconds.)

2. $10^{80}+127$. The website will try ECM and it won't find factors. Then it will run the quadratic sieve. (This examples takes about 6 minutes on 20 seconds to run on a fast computer.)